

Mathematics for Data Science & Machine Learning

Comprehensive Practical Question Bank: 200 Core Applied Problems

This document serves as an exhaustive problem-solving workbook containing exactly 200 meticulously designed questions across four pillars of machine learning mathematics: Linear Algebra, Probability Theory, Statistics, and Differential Calculus. The questions step progressively from foundational mechanics to complex algorithmic applications (e.g., Backpropagation, PCA, OLS Regression, and Maximum Likelihood Estimation). Practice working through these analytically and numerically to anchor your mathematical intuition for ML models.

1. Linear Algebra (Questions 1–50)

A. FOUNDATIONAL & BASIC LEVEL (QUESTIONS 1–15)

- Given vectors $\mathbf{u} = [3, -1, 4]^T$ and $\mathbf{v} = [-2, 5, 2]^T$, calculate the linear combination $2\mathbf{u} - 3\mathbf{v}$ and compute its Euclidean (L_2) norm.
- Compute the dot product of $\mathbf{x} = [1, 2, 0, -3]$ and $\mathbf{y} = [4, -1, 5, 2]$. Are these two vectors orthogonal? Justify mathematically.
- Find the Manhattan (L_1) norm and the Chebyshev (L_∞) norm of the weight error vector $\mathbf{e} = [-0.5, 1.2, 0.0, -2.1, 0.8]$.
- Calculate the cosine similarity between an evaluation query vector $\mathbf{q} = [1, 3, 2]$ and a document text vector $\mathbf{d} = [2, 0, 4]$.
- Compute the matrix-vector product $\mathbf{A}\mathbf{x}$ where:

$$\begin{aligned} \mathbf{A} &= \begin{bmatrix} 2 & -1 & 3 \\ 0 & 4 & 5 \\ -1 & 2 & 1 \end{bmatrix} \\ \mathbf{x} &= [2, 1, -3]^T \end{aligned}$$

- For the matrices $\mathbf{A}$ and $\mathbf{B}$ below, compute the full matrix product $\mathbf{C} = \mathbf{AB}$:

$$\begin{aligned} \mathbf{A} &= \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix} \\ \mathbf{B} &= \begin{bmatrix} 0 & -1 & 2 \\ 1 & 3 & -2 \end{bmatrix} \end{aligned}$$

- Verify if matrix multiplication is commutative for the following pair by evaluating $\mathbf{AB}$ and $\mathbf{BA}$:

$$\begin{aligned} \mathbf{A} &= \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \\ \mathbf{B} &= \begin{bmatrix} 1 & -1 \\ 2 & 4 \end{bmatrix} \end{aligned}$$

8. Compute the transpose of the matrix product $(AB)^T$ and prove it equals $B^T A^T$ using:

$$A = \begin{bmatrix} 1 & 3 \\ 2 & 0 \end{bmatrix}$$
$$B = \begin{bmatrix} -1 & 4 \\ 2 & 3 \end{bmatrix}$$

9. Find the determinant of the 2x2 data scaling matrix $A = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}$. Is this matrix invertible?

10. Compute the explicit inverse A^{-1} for the matrix:

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$$

11. Calculate the trace of the square feature covariance matrix Σ given by:

$$\Sigma = \begin{bmatrix} 12.5 & 4.2 & -1.1 \\ 4.2 & 8.3 & 0.5 \\ -1.1 & 0.5 & 15.1 \end{bmatrix}$$

12. Determine if the vector set $\mathbf{v}_1 = [1, 2, 1]$, $\mathbf{v}_2 = [2, 9, 0]$, and $\mathbf{v}_3 = [3, 11, 1]$ is linearly independent by setting up a homogeneous linear system.

13. Identify the rank of the following matrix by reducing it to its row echelon form (REF):

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 5 & 7 \end{bmatrix}$$

14. Project the feature vector $\mathbf{x} = [4, 5]^T$ onto the line spanned by the unit direction vector $\mathbf{u} = [0.6, 0.8]^T$. Find the scalar projection and the vector projection.

15. Determine whether the transformation matrix Q is orthogonal:

$$Q = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

B. INTERMEDIATE LEVEL (QUESTIONS 16–35)

16. Solve the linear system $A\mathbf{x} = \mathbf{b}$ via Gaussian Elimination, representing a 3-feature equilibrium model:

$$\begin{aligned} 2x + y - z &= 8 \\ -3x - y + 2z &= -11 \\ -2x + y + 2z &= -3 \end{aligned}$$

17. Compute the determinant of the 3x3 transformation matrix using cofactor expansion:

$$A = \begin{bmatrix} 1 & 3 & 2 \\ 4 & -1 & 5 \\ 2 & 3 & 1 \end{bmatrix}$$

18. Find the inverse of the 3x3 matrix M using the Gauss-Jordan elimination method:

$$M = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 3 \\ 0 & 1 & 1 \end{bmatrix}$$

19. Find the eigenvalues and corresponding eigenvectors for the 2x2 transformation matrix $A = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}$.

20. For the symmetric matrix $S = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, find its orthogonal diagonalization components P and D such that $S = PDP^{-1}$.

21. Compute the Frobenius norm of the given weight error matrix W :

$$W = \begin{bmatrix} 0.5 & -1.2 \\ -2.0 & 0.8 \\ 1.1 & 0.0 \end{bmatrix}$$

22. Given a dataset matrix X of 3 samples with 2 features, compute the uncentered scatter matrix $S = X^T X$:

$$X = \begin{bmatrix} 1 & 2 \\ 3 & 0 \\ 2 & 4 \end{bmatrix}$$

23. Determine if the quadratic form $f(x) = 3x_1^2 + 2x_1x_2 + 3x_2^2$ is positive definite by checking the eigenvalues or principal minors of its characteristic matrix.

24. Transform the basis of vector $x_B = [2, -1]^T$ from basis $B = \{[1, 1]^T, [0, 2]^T\}$ into the standard standard Euclidean basis.

25. Compute the Hadamard (element-wise) product $A \odot B$ and the standard matrix product AB for the following filters:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \\ B = \begin{bmatrix} 0 & 5 \\ 2 & 1 \end{bmatrix}$$

26. Using the Gram-Schmidt process, orthogonalize the linearly independent vectors $v_1 = [1, 1, 0]$ and $v_2 = [1, 0, 1]$ to form an orthonormal set.

27. Compute the null space (kernel) of the matrix A by solving $Ax = 0$:

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 3 & 6 & 3 \end{bmatrix}$$

28. Find the condition number $\kappa(A)$ using the L_1 norm for the poorly conditioned matrix:

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 1.001 \end{bmatrix}$$

29. Decompose the matrix $A = \begin{bmatrix} 4 & 3 \\ 6 & 3 \end{bmatrix}$ into an upper triangular matrix U and a lower triangular matrix L such that $A = LU$ (LU Decomposition).

30. Given a graph with adjacency matrix $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$, calculate A^3 to determine the number of paths of length 3 between node 1 and node 2.

31. Compute the column space and row space bases for the matrix:

$$M = \begin{bmatrix} 1 & 4 & 2 \\ 2 & 8 & 4 \\ 0 & 1 & 3 \end{bmatrix}$$

32. Calculate the rank-deficient pseudo-inverse (Moore-Penrose) A^+ of the vector $A = [1, 2, -1]^T$.

33. Express the vector $w = [7, 11]$ as a linear combination of the basis vectors $u_1 = [2, 3]$ and $u_2 = [1, 2]$.

34. An affine transformation is given by $f(x) = Ax + b$. Compute the transformed coordinates of $x = [1, 2]^T$ given:

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \quad (90^\circ \text{ degree rotation})$$

$$b = [3, -1]^T \quad (\text{translation})$$

35. Show that the matrix $P = A(A^T A)^{-1} A^T$ is idempotent by verifying that $P^2 = P$. What is the physical meaning of this matrix in linear regression?

C. ADVANCED & ML-APPLICATION LEVEL (QUESTIONS 36–50)

36. Find the singular value decomposition (SVD) $A = U\Sigma V^T$ for the rectangular matrix:

$$A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}$$

37. Given the data matrix X consisting of 3 samples with 2 features (centered at zero):

$$X = \begin{bmatrix} -1 & -1 \\ 0 & 2 \\ 1 & -1 \end{bmatrix}$$

Compute the empirical covariance matrix $\Sigma = (1/n)X^T X$ and find the principal component directions (eigenvectors).

38. Solve the overdetermined system of equations $Xw = y$ via the normal equation $w = (X^T X)^{-1} X^T y$ for the linear regression setup:

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$$

$$y = [2, 3, 5]^T$$

39. Apply spectral decomposition to project the data point $x = [2, 3]$ onto the first principal component of the covariance matrix $\Sigma = \begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix}$.

40. Compute the low-rank approximation A_1 (Rank-1 approximation via SVD) of the matrix:

$$A = \begin{bmatrix} 5 & 0 \\ 0 & 1 \end{bmatrix}$$

41. Compute the Mahalanobis distance between data point $x = [2, 4]^T$ and the distribution mean $\mu = [0, 0]^T$ given the inverse covariance matrix:

$$\Sigma^{-1} = \begin{bmatrix} 0.5 & -0.2 \\ -0.2 & 0.3 \end{bmatrix}$$

42. In a Graph Convolutional Network, you have the adjacency matrix A and degree matrix D . Compute the symmetric normalized Laplacian $L_{sym} = I - D^{-1/2}AD^{-1/2}$ for:

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

43. Derive the gradient of the ridge regression regularization term $f(w) = \lambda \|w\|_2^2 = \lambda w^T w$ with respect to the weight vector w .

44. Find the explicit pseudo-inverse $X^+ = (X^T X)^{-1} X^T$ for the non-square data matrix:

$$X = \begin{bmatrix} 1 & 2 \\ 2 & 1 \\ 1 & 1 \end{bmatrix}$$

45. Show that if λ is an eigenvalue of an orthogonal matrix Q , then $|\lambda| = 1$. Prove this using the property $Q^T Q = I$.

46. A Markov transition matrix is given by $P = \begin{bmatrix} 0.8 & 0.2 \\ 0.3 & 0.7 \end{bmatrix}$. Find the steady-state vector π satisfying $\pi P = \pi$ subject to $\pi_1 + \pi_2 = 1$.

47. Using the kernel trick, map two 1D points $x_1 = 2, x_2 = 5$ into a 3D space using the polynomial kernel feature map $\Phi(x) = [x^2, \mu 2x, 1]^T$. Show that $\Phi(x_1)^T \Phi(x_2) = (x_1 x_2 + 1)^2$.

48. Compute the transformation matrix required to white (whiten) a dataset whose covariance matrix is $\Sigma = \begin{bmatrix} 4 & 0 \\ 0 & 9 \end{bmatrix}$.

49. Evaluate the tensor outer product $A \otimes B$ for vectors $A = [1, 2]$ and $B = [3, 4, 5]$, demonstrating how cross-feature combinations are built.

50. Given a deep neural network layer with weight matrix W , express the condition required to prevent vanishing and exploding gradients in terms of the singular values of W .

2. Probability (Questions 51–100)

A. FOUNDATIONAL & BASIC LEVEL (QUESTIONS 51–65)

51. A server room contains 60% Linux servers and 40% Windows servers. If 5% of Linux servers and 10% of Windows servers experience downtime today, what is the total probability that a randomly chosen server experiences downtime?
52. A sample space contains events A and B where $P(A) = 0.4$, $P(B) = 0.5$, and $P(A \cup B) = 0.7$. Calculate the conditional probability $P(A | B)$. Are A and B independent?
53. An email spam filter spots words based on a dictionary. 8% of all emails are spam. The word "FREE" appears in 20% of spam emails and in 2% of non-spam emails. Use Bayes' Theorem to calculate $P(\text{Spam} | \text{"FREE"})$.
54. A discrete random variable X has the following probability mass function (PMF):
- | | | | | |
|---------|------|------|------|------|
| $X:$ | 1, | 2, | 3, | 4] |
| $P(X):$ | 0.1, | 0.4, | 0.3, | 0.2] |
- Calculate the expected value $E[X]$.
- 55.
- For the discrete random variable X defined in Question 54, calculate its variance $\text{Var}(X)$ and standard deviation σ .
56. Compute the probability of getting exactly 7 heads in 10 independent flips of a fair coin using the Binomial Distribution formula.
57. The number of API requests hitting a microservice follows a Poisson distribution with an average of 4 requests per second. Find the probability that exactly 2 requests arrive in any given second.
58. A continuous random variable X is uniformly distributed between 10 and 50. Calculate $P(20 \leq X \leq 35)$ and find its cumulative distribution function (CDF) for $x = 30$.
59. Assume standard user session times are normally distributed with $\mu = 5$ minutes and $\sigma = 1.5$ minutes. Convert a session time of $x = 8$ minutes into a standard Z-score.
60. Using a standard normal distribution table (where $\Phi(2.0) = 0.9772$), find the probability that a session lasts less than 8 minutes for the distribution described in Question 59.
61. The probability of a user clicking an ad is $p = 0.02$. What is the expected number of ad impressions required before the first click occurs (Geometric Distribution)?
62. If events A , B , and C are mutually exclusive with probabilities 0.2, 0.35, and 0.15 respectively, find the probability that *none* of these events occur.
63. A binary classification model makes predictions on a test set. Define the sample space elements for outcomes in terms of True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN). Express Precision as a conditional probability.
64. A database connection has a failure rate of 1% per query. What is the probability that the first failure happens on exactly the 5th query?

65. Prove that for any two events A and B , $P(A \cap B) \leq P(A) \leq P(A \cup B)$.

B. INTERMEDIATE LEVEL (QUESTIONS 66–85)

66. A diagnostic test for a disease has a sensitivity of 99% and a specificity of 95%. The disease is rare, affecting only 0.5% of the population. If a patient tests positive, what is the probability they actually have the disease?

67. Two discrete random variables X and Y have the following joint PMF:

	$Y=0$	$Y=1$
$X=0$	0.2	0.1
$X=1$	0.3	0.4

Compute the marginal distributions $P(X)$ and $P(Y)$.

68. For the joint distribution in Question 67, calculate the conditional expectation $E[Y | X=1]$.

69. Calculate the exact covariance $\text{Cov}(X, Y)$ for the joint distribution provided in Question 67.

70. Using the covariance from Question 69 and the marginal variances, calculate the Pearson correlation coefficient $\rho_{X,Y}$.

71. The lifetime of a hard drive follows an exponential distribution with a mean time to failure (MTTF) of 10,000 hours (rate parameter $\lambda = 0.0001$). Find the probability that a drive lasts more than 15,000 hours.

72. Given independent random variables X and Y with $\text{Var}(X) = 4$ and $\text{Var}(Y) = 9$, find the variance of the combined linear scaling feature $W = 3X - 2Y$.

73. The continuous random variable X has a probability density function (PDF) given by $f(x) = cx^2$ for $0 \leq x \leq 2$, and 0 otherwise. Find the normalization constant c .

74. For the PDF defined in Question 73 with the corrected constant c , calculate the expected value $E[X]$.

75. Find the cumulative distribution function (CDF), $F(x)$, for the continuous random variable in Question 73.

76. A machine learning model makes errors that follow a Poisson process with an average rate of 0.5 errors per hour. What is the probability of seeing 0 errors over a 4-hour evaluation window?

77. A system relies on two independent parallel components. Component A has a reliability of 0.95 and Component B has a reliability of 0.90. What is the overall probability that the system functions?

78. Use the Central Limit Theorem to approximate the probability that the sum of 50 independent uniform random variables (each distributed between 0 and 1) is greater than 27. (Note: For a single uniform $U(0,1)$, $\mu=0.5$, $\sigma^2=1/12$).

79. Prove analytically that $\text{Var}(X) = E[X^2] - (E[X])^2$ using the linearity properties of expectation.

80. Let $X \sim N(10, 4)$ and $Y \sim N(20, 9)$ be independent feature variables. Find the distribution parameter specifications for the new combined feature vector $Z = X + Y$.

81. If a student guesses randomly on a 5-question multiple-choice quiz where each question has 4 options, what is the probability they get at least 4 questions correct?

82. Suppose the conditional probability $P(\text{Feature} \mid \text{Class } 1) = 0.8$ and $P(\text{Feature} \mid \text{Class } 0) = 0.2$. Assuming uniform prior class probabilities, find the posterior probability $P(\text{Class } 1 \mid \text{Feature})$.
83. A container holds 5 red balls and 5 blue balls. If you draw 3 balls without replacement, find the probability of getting exactly 2 red balls using the Hypergeometric distribution.
84. Show that the entropy $H(X) = -\sum p(x) \log_2 p(x)$ for a fair coin flip is exactly 1 bit, while a highly biased coin with $p=0.99$ yields significantly lower entropy.
85. State Chebyshev's Inequality. If a distribution has mean $\mu = 100$ and standard deviation $\sigma = 10$, provide a lower bound on the probability that a sample falls between 80 and 120.

C. ADVANCED & ML-APPLICATION LEVEL (QUESTIONS 86–100)

86. Consider a bivariate normal distribution modeling two features X_1 and X_2 with mean vector $\mu = [0, 0]^T$ and covariance matrix $\Sigma = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix}$. Write out the explicit algebraic form of the joint density function $f(x_1, x_2)$.
87. For the bivariate normal distribution in Question 86, derive the conditional distribution of X_1 given that $X_2 = 1.5$ (i.e., find $\mu_{1|2}$ and $\sigma_{1|2}^2$).
88. Define the Kullback-Leibler (KL) Divergence $D_{KL}(P \parallel Q)$ for discrete distributions. Calculate $D_{KL}(P \parallel Q)$ where $P = [0.6, 0.4]$ and $Q = [0.5, 0.5]$.
89. Show that the KL divergence is non-negative, $D_{KL}(P \parallel Q) \geq 0$, and explain why it cannot serve as a true distance metric in clustering algorithms.
90. In a Naive Bayes classifier, given class label C and conditionally independent features $X_1, X_2, \dots, X_n$, write out the explicit proportional equation used to compute the posterior class probability.
91. Let $X_1, X_2, \dots, X_n$ be independent and identically distributed (i.i.d.) random variables following a Bernoulli distribution with parameter p . Derive the Moment Generating Function (MGF) for a single variable X_1 .
92. Use the pseudo-random transformation technique (Inverse CDF Method) to derive a formula for generating samples from an Exponential distribution with parameter λ given uniform samples $U \sim \text{Uniform}(0,1)$.
93. A Gaussian Mixture Model (GMM) combines two Gaussians with mixing coefficients $\pi_1 = 0.3$ and $\pi_2 = 0.7$. If $p_1(x) = N(x; 0, 1)$ and $p_2(x) = N(x; 5, 2)$, write the complete density function expression $p(x)$.
94. Derive the formula for the expectation of a quadratic form $E[X^T A X]$ where X is a random vector with mean μ and covariance matrix Σ .
95. Explain the concept of a conjugate prior in Bayesian inference. If the likelihood follows a Binomial distribution, what is the mathematically convenient conjugate prior distribution for the parameter p ?

96. A Markov chain page-rank model has three states. The transition matrix is:

$$P = \begin{bmatrix} 0.1 & 0.6 & 0.3 \\ 0.4 & 0.2 & 0.4 \\ 0.2 & 0.3 & 0.5 \end{bmatrix}$$

Given an initial state probability distribution $\mathbf{v}_0 = [1.0, 0, 0]$, compute the state probabilities after 2 transitions ($\mathbf{v}_2$).

97. Use Hoeffding's Inequality to determine the maximum probability that the empirical mean error of a machine learning model deviates from its true expected error by more than $\epsilon = 0.05$ given $n = 1000$ test samples bounded within $[0, 1]$.

98. Write out the joint distribution definition for a Dirichlet distribution over a probability simplex with parameters $\alpha = [2, 2, 2]$. How is this distribution applied in Latent Dirichlet Allocation (LDA) topic modeling?

99. Using Jensen's Inequality, prove that the log-likelihood of a mixture model satisfies the lower bound criteria used during the Expectation-Maximization (EM) algorithm update step.

100. Let X and Y be independent standard normal random variables. Define a new variable $Z = X/Y$. Identify the resulting distribution type of Z and discuss its mean and variance properties.

3. Statistics (Questions 101–150)

A. FOUNDATIONAL & BASIC LEVEL (QUESTIONS 101–115)

101. Calculate the sample mean ($\bar{x}$) and median for the following list of model validation accuracy scores:

Data: $[0.85, 0.92, 0.88, 0.76, 0.94, 0.89, 0.91]$

102. Compute the sample variance (s^2) and sample standard deviation (s) for the model scores provided in Question 101. Use Bessels Correction ($n - 1$).

103. Find the range and the Interquartile Range (IQR) for the given dataset representing website click latency (in milliseconds):

Data: $[45, 52, 61, 68, 74, 85, 99, 120, 250]$

104. Identify any outliers in the latency dataset of Question 103 using the standard $1.5 \times \text{IQR}$ rule.

105. A sample of size $n = 64$ has a sample mean $\bar{x} = 105$. Assuming a known population standard deviation $\sigma = 16$, calculate the 95% confidence interval for the population mean μ (use $Z_{0.025} = 1.96$).

106. State the clear difference between the Null Hypothesis (H_0) and the Alternative Hypothesis (H_1) in the context of testing if a new model outperforms an old baseline model.

107. Define what Type I Error (α) and Type II Error (β) represent when deploying an anomaly detection classification model.

108. A marketing team runs an A/B test and arrives at a p-value of 0.034. Explain what this p-value actually means relative to the null hypothesis. Do you reject H_0 at a significance level of $\alpha = 0.05$?

109. Compute the standard error of the mean (SEM) for a feature sample where $s = 12.4$ and the sample size is $n = 100$.

110. Given the following observed counts of user preferences across three UI themes, calculate the expected counts under the hypothesis that all three themes are equally preferred:

Theme A: 45, Theme B: 35, Theme C: 40

111. Calculate the sample skewness value qualitatively for a dataset where **Mean** = 45, **Median** = 52, and **Mode** = 58. Is the distribution skewed left or right?

112. Convert the following raw data vectors into a single 2x2 matrix format to prepare for calculating sample covariance:

Feature X: [2, 4, 6]

Feature Y: [5, 7, 9]

113. Explain what the Central Limit Theorem (CLT) implies about the shape of the sampling distribution of the sample mean as $n \rightarrow \infty$, even if the underlying population distribution is heavily skewed.

114. An analyst samples 1000 customer ratings and finds a 99% confidence interval of [4.2, 4.6]. Does this mean 99% of all individual customer ratings fall between 4.2 and 4.6? Clarify the misconception.

115. Define the term "Degrees of Freedom" in the context of estimating a sample variance from n data points.

B. INTERMEDIATE LEVEL (QUESTIONS 116–135)

116. Perform a one-sample t-test manually. A deep learning model's average inference latency is claimed to be 50ms. You run $n = 9$ trials and find $\bar{x} = 54ms$ and $s = 6ms$. Compute the t-statistic. At $df = 8$, $t_{critical} = 2.306$ for $\alpha = 0.05$ (two-tailed); state your conclusion.

117. You are comparing two models (A and B) via an independent two-sample t-test with equal variances. Given the following summary statistics, compute the pooled variance s_p^2 and the final t-statistic:

Model A: $n_1 = 15$, $\bar{x}_1 = 0.88$, $s_1 = 0.04$

Model B: $n_2 = 15$, $\bar{x}_2 = 0.85$, $s_2 = 0.05$

118. Construct a 95% confidence interval for the true difference between the two model means using the data from Question 117 (assume $t_{crit} \approx 2.048$).

119. Calculate the Chi-Square (χ^2) statistic for the following 2x2 contingency table evaluating user retention across two distinct application versions:

Version	Retained	Churned
Version A	120	80
Version B	150	50

120. Given the Chi-Square statistic from Question 119, determine the degrees of freedom and state whether the version type and retention are independent at $\alpha = 0.05$ (Critical χ^2 for $df=1$ is **3.841**).

121. Compute the exact Pearson correlation coefficient r between the independent input variable X and target variable Y from this small training sample:

```
X = [1, 2, 3, 4, 5]
Y = [2, 5, 5, 8, 10]
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122. Fit a simple ordinary least squares (OLS) linear regression line $y = \beta_0 + \beta_1 x$ using the data from Question 121 by computing the explicit slope $\beta_1 = \text{Cov}(X,Y)/\text{Var}(X)$ and intercept β_0 .

123. Using the linear regression model derived in Question 122, compute the predicted values $\hat{y}$ and find the Total Sum of Squares (SST) and Residual Sum of Squares (SSR). Then, evaluate the Coefficient of Determination R^2 .

124. Explain the concept of statistical power ($1 - \beta$). List three specific methods an ML engineer can employ to increase the power of an online web banner evaluation experiment.

125. An algorithmic trading bot shows a daily return series with mean 0.1% and standard deviation 0.2%. Assuming normality, calculate the daily Value at Risk (VaR) at a 95% confidence level ($Z = -1.645$).

126. Convert the following raw ranking positions into a Spearman's rank correlation coefficient to handle non-linear monotonic data trends:

```
Rank X: [1, 2, 3, 4, 5]
Rank Y: [2, 1, 4, 3, 5]
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127. Explain the difference between a parametric test and a non-parametric test. When should you prefer the Mann-Whitney U test over an independent two-sample t-test?

128. Set up a one-way ANOVA table framework. You are testing three feature engineering methods across 4 model iterations each. Total Sum of Squares (SST) = 150, and Between-Group Sum of Squares (SSB) = 90. Compute the Within-Group Sum of Squares (SSW) and the final F-statistic.

129. Define Cohen's d for measure of effect size. Calculate Cohen's d for the model metrics presented in Question 117 to evaluate practical significance versus mere statistical significance.

130. A sample size of $n = 16$ yields a sample mean of 50 and a sample standard deviation of 8. Find the 95% confidence interval for μ using the student's t-distribution score ($t_{0.025, df=15} = 2.131$).

131. Describe the exact difference between the standard normal distribution (Z) and the Student's t-distribution. Under what conditional constraints do they converge?

132. Suppose you perform 20 independent hypothesis tests simultaneously at $\alpha = 0.05$. What is the probability of making at least one Type I error by pure chance? Apply the Bonferroni Correction to find the adjusted significance threshold.

133. Calculate the F-statistic denominator and numerator degrees of freedom for an ANOVA design containing $k = 4$ comparison groups and a total of $N = 40$ observations.

134. Define the term "unbiased estimator." Prove that the sample mean $\bar{x}$ is an unbiased estimator of the population mean μ .

135. Write down the probability density function for the Chi-Square distribution with k degrees of freedom and explain its relationship to standard normal variables.

C. ADVANCED & ML-APPLICATION LEVEL (QUESTIONS 136–150)

136. Let $x_1, x_2, \dots, x_n$ be i.i.d. random samples drawn from an Exponential distribution with parameter λ . Derive the Maximum Likelihood Estimator (MLE) formula for λ by constructing the log-likelihood function and setting its derivative to zero.

137. Given a sample dataset of server times to failure in hours: $[100, 200, 150, 250]$. Calculate the concrete numerical MLE estimate for the parameter λ based on your derivation in Question 136.

138. Derive the Maximum Likelihood Estimator for the mean μ of a normal distribution $N(\mu, \sigma^2)$ given n i.i.d. observations.

139. Derive the Maximum A Posteriori (MAP) estimator for a Bernoulli parameter p using a Beta distribution prior with hyperparameters α and β . Show how MAP acts as a regularized version of MLE.

140. Explain the Bias-Variance Decomposition of the Mean Squared Error (MSE). Write out the complete equation breaking down MSE into Bias square, Variance, and Irreducible Error.

141. A model has an expected prediction curve. If its true function is $f(x) = x^2$, its average estimated model function across multiple training sets is $g(x) = x^2 + 0.5$, and the noise variance is $\sigma^2 = 0.1$, compute the total MSE at any point x .

142. Explain how the Bootstrap resampling method works to estimate the standard error of a non-analytical metric like the median score of a complex model. Outline the step-by-step loop process.

143. Compute the log-odds and the corresponding probability value for a logistic regression model output given the weight vector $w = [0.5, -1.2]^T$, bias $b = 0.2$, and input feature vector $x = [2.0, 1.0]^T$.

144. Derive the Fisher Information matrix for a single parameter Bernoulli trial model. How does this relate to the variance of the MLE estimator via the Cramér-Rao Lower Bound?

145. Explain the difference between Ridge (L_2) and Lasso (L_1) regression from a Bayesian perspective by identifying the corresponding prior distributions placed on the weight vectors.

146. In multiple linear regression modeling, explain what Multicollinearity is. How do you use the Variance Inflation Factor (VIF) to diagnose it?

147. Define the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC). Calculate both metrics for a model with $k = 5$ parameters, log-likelihood $\ln L = -42.5$, and sample size $n = 100$.

148. Explain the concept of standard cross-validation (e.g., K-fold CV). Why does evaluating model accuracy exclusively on the training set yield a biased statistic?

- 149.** A linear regression model violates the homoscedasticity assumption. Explain what heteroscedasticity is and describe its structural impact on calculated p-values and confidence intervals.
- 150.** What is the False Discovery Rate (FDR)? How does the Benjamini-Hochberg procedure differ fundamentally from the Bonferroni correction when handling thousands of feature selection tests?

4. Differential Calculus (Questions 151–200)

A. FOUNDATIONAL & BASIC LEVEL (QUESTIONS 151–165)

- 151.** Compute the first derivative of the polynomial activation feature function: $f(x) = 4x^3 - 5x^2 + 2x - 7$.
- 152.** Find the derivative of the natural log function scaling term: $f(x) = \ln(3x)$ for $x > 0$.
- 153.** Apply the Product Rule to find the derivative of the combined function: $f(x) = x^2 e^x$.
- 154.** Apply the Quotient Rule to find the derivative of the function: $f(x) = x / (x + 1)$.
- 155.** Using the Chain Rule, find the explicit derivative of the function: $f(x) = (3x^2 + 1)^4$.
- 156.** Find the slope of the tangent line to the curve $f(x) = 2x^2 - 3x + 1$ at the specific data coordinate point $x = 2$.
- 157.** Evaluate the first derivative of the classic Sigmoid activation function $\sigma(x) = 1 / (1 + e^{-x})$. Express your final answer strictly in terms of $\sigma(x)$.
- 158.** Find the partial derivatives $\partial f / \partial x$ and $\partial f / \partial y$ for the multi-variable function: $f(x, y) = 3x^2y + 5xy^3$.
- 159.** Compute the gradient vector $\nabla f(x, y)$ for the function $f(x, y) = 2x^3 - 4xy + y^2$ at the specific operational point $(1, 2)$.
- 160.** Identify all critical points of the function $f(x) = x^2 - 4x + 5$ by setting the derivative to zero. Determine whether it represents a local minimum or maximum.
- 161.** Compute the second derivative $f''(x)$ of the loss function $f(x) = x^4 - 6x^2$. Find the points of inflection where $f''(x) = 0$.
- 162.** Find the derivative of the Rectified Linear Unit (ReLU) activation function, $f(x) = \max(0, x)$. Discuss its value and continuity at exactly $x = 0$.
- 163.** Apply L'Hôpital's Rule to evaluate the numerical limit of the indeterminate ratio: $\lim_{(x \rightarrow 0)} (\sin(x) / x)$.
- 164.** Compute the directional derivative of $f(x, y) = x^2 + 2xy$ at the point $(1, 1)$ in the direction of the unit vector $u = [0.6, 0.8]^T$.
- 165.** Write out the first-order Taylor series approximation formula for a general function $f(x)$ centered around a localized pivot point $x = a$.

B. INTERMEDIATE LEVEL (QUESTIONS 166–185)

- 166.** Find the gradient vector ∇f for the Mean Squared Error loss function for a single training sample: $f(\mathbf{w}_1, \mathbf{w}_2) = (w_1 x_1 + w_2 x_2 - y)^2$ with respect to vector $\mathbf{w} = [w_1, w_2]$.
- 167.** Evaluate the gradient of the function from Question 166 using the real data values: $x_1 = 2, x_2 = -1, y = 3$, evaluated at current weights $w_1 = 1, w_2 = 1$.
- 168.** Perform one explicit numerical step of Gradient Descent to update the weight vector $\mathbf{w}$. Use the current weights and gradient vector from Question 167 with a learning rate parameter $\eta = 0.1$ via $\mathbf{w}^{new} = \mathbf{w}^{old} - \eta \nabla f$.
- 169.** Compute the 2x2 Hessian matrix $\mathbf{H}$ of second-order partial derivatives for the multi-variable optimization profile: $f(x, y) = 3x^2 - 4xy + 2y^2$.
- 170.** Analyze the eigenvalues of the Hessian matrix calculated in Question 169 to determine whether the critical point at $(0,0)$ is a local minimum, local maximum, or a saddle point.
- 171.** Find the derivative of the Hyperbolic Tangent ($\tanh$) activation function, $f(x) = (e^x - e^{-x}) / (e^x + e^{-x})$. Express your solution in terms of $\tanh(x)$.
- 172.** Compute the derivative of the binary cross-entropy loss function for a single sample: $L(y, \hat{y}) = -(y \ln(\hat{y}) + (1-y) \ln(1-\hat{y}))$ with respect to the predicted probability $\hat{y}$.
- 173.** Using the chain rule, compute the derivative of the binary cross-entropy loss L with respect to the pre-activation logit z , given that $\hat{y} = \sigma(z)$. Combine answers from Questions 157 and 172.
- 174.** Find the second-order Taylor series approximation for the function $f(x) = \ln(x)$ centered around the evaluation pivot $a = 1$.
- 175.** Use the concept of implicit differentiation to find the slope dy/dx of the implicit regularized circle boundary curve: $x^2 + y^2 = 25$ at the exact coordinate position $(3, 4)$.
- 176.** Compute the Jacobian matrix $\mathbf{J}$ for a vector-valued transformation function $F(x, y) = [x^2 + y, 3x - y^2]^T$.
- 177.** Evaluate the Jacobian matrix $\mathbf{J}$ from Question 176 numerically at the sample coordinates $(x=2, y=3)$.
- 178.** Find the global minimum of the regularized cost curve $J(\mathbf{w}) = (\mathbf{w} - 4)^2 + 2\mathbf{w}^2$ by optimizing analytically.
- 179.** Given the scalar function of a vector $f(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x}$ where $\mathbf{A}$ is a non-symmetric square matrix, write the general matrix derivative expression for $\nabla_{\mathbf{x}} f(\mathbf{x})$.
- 180.** Explain the mathematical definition of a convex function in terms of its Hessian matrix. What condition must the Hessian satisfy across its domain?
- 181.** Compute the derivative of the Leaky ReLU activation function: $f(x) = \max(0.01x, x)$ for both $x > 0$ and $x < 0$.
- 182.** Find the partial derivative of the Softplus activation function $f(x) = \ln(1 + e^x)$. Show that its derivative simplifies exactly to the standard Sigmoid function.

- 183.** Determine the critical points of the function $f(x, y) = x^3 - 3x + y^2$ and classify them using the Second Derivative Test (Hessian determinant criterion).
- 184.** Explain how the learning rate η influences convergence in gradient descent. What happens mathematically if the learning rate exceeds the maximum eigenvalue of the system's Hessian?
- 185.** Compute the partial derivative of the log-sum-exp function $f(x_1, x_2) = \ln(e^{x_1} + e^{x_2})$ with respect to x_1 .

C. ADVANCED & ML-APPLICATION LEVEL (QUESTIONS 186–200)

- 186.** Using the method of Lagrange Multipliers, minimize the function $f(x, y) = x^2 + y^2$ subject to the hard linear equality constraint $x + y = 2$. Find the optimal values for x , y , and the Lagrange multiplier λ .
- 187.** Consider a single neuron with input x , weight w , bias b , pre-activation $z = wx + b$, and activation $a = \sigma(z)$. If the cost is $C = 0.5(a - y)^2$, use the Chain Rule to derive the explicit backpropagation error gradient formulas $\partial C / \partial w$ and $\partial C / \partial b$.
- 188.** Given the multi-class Softmax activation function for an output vector z : $a_i = e^{z_i} / \sum_j e^{z_j}$. Derive the partial derivative $\partial a_i / \partial z_j$ for the case where $i = j$.
- 189.** Derive the partial derivative $\partial a_i / \partial z_j$ for the Softmax function from Question 188 for the case where $i \neq j$.
- 190.** Derive the full vector-matrix gradient of the multivariate OLS objective function $L(w) = \|Xw - y\|_2^2 = (Xw - y)^T (Xw - y)$ with respect to the parameter weight vector w . Show each matrix expansion step.
- 191.** Set the matrix gradient derived in Question 190 to zero to analytically prove the structure of the standard Normal Equation: $X^T X w = X^T y$.
- 192.** Explain the mechanics of Newton-Raphson optimization for a multi-variable function. Write out the explicit weight update equation involving both the gradient vector and the inverse Hessian matrix: $w^{new} = w^{old} - H^{-1} \nabla f$.
- 193.** Apply the matrix derivative rule to find the gradient of the log-determinant of a positive-definite covariance matrix, $f(\Sigma) = \ln|\Sigma|$, with respect to Σ .
- 194.** Write out the Karush-Kuhn-Tucker (KKT) conditions for an optimization problem containing inequality constraints, such as those found in Support Vector Machine (SVM) max-margin boundary formulas.
- 195.** Derive the gradient of the standard Frobenius norm squared of a matrix weight parameter, $f(W) = \|W\|_F^2$, with respect to the matrix W itself.
- 196.** In a recurrent neural network (RNN), hidden states update via $h_t = \tanh(W_{hh}h_{t-1} + W_{xh}x_t)$. Express the derivative formulation of $\partial h_t / \partial h_{t-1}$ to highlight why long sequences trigger vanishing gradients.
- 197.** Find the total differential df for a neural network node with structural interactions: $f(x, y) = x^2 e^{-y}$.
- 198.** Given the vector function $f(x) = \ln(w^T x)$, find the gradient with respect to the parameter vector w .
- 199.** Show that the negative gradient direction $-\nabla f(x)$ is the path of steepest local descent by optimizing the directional derivative constraint.

200. For a complex regularized neural network loss function $E(\mathbf{w}) = L(\mathbf{w}) + (\lambda/2)\mathbf{w}^T\mathbf{w}$, find the second-order derivative (Hessian) with respect to $\mathbf{w}$, given that the Hessian of the unregularized loss $L(\mathbf{w})$ is $\mathbf{H}_L$.